Geoderma 455 (2025) 117197

Contents lists available at ScienceDirect

Geoderma

journal homepage: www.elsevier.com/locate/geoderma

Edaphic and climatic effects on soil water dynamics and infiltration patterns in tropical rainforests

Xin Zou[a,b,c,d], Ankit Shekhar[c], Yuxuan Mo[a], Ashutosh Kumar Singh[a], Xiaojin Jiang[a], Wenjie Liu[a,*]

[a] CAS Key Laboratory of Tropical Forest Ecology, Xishuangbanna Tropical Botanical Garden, Chinese Academy of Sciences, Menglun, Yunnan 666303, China
[b] University of Chinese Academy of Sciences, Beijing 100049, China
[c] ETH Zurich, Department of Environmental Systems Science, 8092 Zurich, Switzerland
[d] Ecological Restoration Biodiversity Conservation Key Laboratory of Sichuan Province, Chengdu Institute of Biology, Chinese Academy of Sciences, Chengdu 610213, China

ARTICLE INFO

Handling editor: Haly Neely

Keywords:
Spatiotemporal stability
Preferential flow
Lateral flow
Uniform flow
Soil physical properties
Soil organic matter

ABSTRACT

The preservation of forests' hydrological functions is pivotal for alleviating land degradation and enhancing ecosystem services. However, understanding of the hydrological processes in tropical rainforests remains inadequate, particularly in the context of a changing climate and complex edaphic conditions. This study aims to quantify the spatiotemporal variations of soil water content and infiltration patterns within tropical rainforests located on distinct slope gradients (flat 0°, moderate 10° and steep 30°), and elucidate the effects of climatic and edaphic factors on these hydrological parameters. Based on the one year observation, it was revealed that the moderate-slope site maintained higher soil water content levels and temporal stability (54.32) compared to the flat (41.85) and steep-slope (42.71) sites across the 0 – 110 cm soil depths. This phenomenon is likely attributed to the superior soil condition at the moderate-slope site, characterized by optimal soil physical properties, a rich content of soil organic matter and abundant root biomass. The regression analysis demonstrated that soil physical properties (represented by capillary holding capacity and bulk density) had significant positive effects on soil water throughout the entire year ($R^2 > 0.6$, $p < 0.01$). The effect of sand content on soil water content was found to be significant only during the dry period ($R^2 = 0.48$, $p < 0.05$), while the effects of soil organic matter and fine roots were significant during both transition and wet periods ($R^2 > 0.5$, $p < 0.05$). In response to the increasing precipitation, the moderate-slope site displayed more equilibrium state of infiltration patterns than the flat and steep-slope sites, evident in the weaker correlations between the dye coverage and the lateral, preferential, and uniform flows. This result suggested that moderate-slope sites may hold greater capacity for retaining soil water. Furthermore, the linear mixed-effects models revealed significant slope and precipitation interaction influencing the infiltrating water flows ($p < 0.01$). Therefore, this study provides valuable insights into the ecohydrological processes in rainforests, which are crucial for the effective planning of future forest management in tropical areas.

1. Introduction

Tropical rainforests are recognized as highly productive ecosystems that provide a diverse array of essential services to society. These services encompass hosting immense biodiversity, acting as the largest terrestrial carbon sinks, and regulating the global hydrological cycle (Jung et al., 2021). However, the delivery of these invaluable ecosystem services is under threat due to unprecedented anthropogenic activities. Deforestation, agriculture expansion, and climate change impacts are major factors that pose significant challenges to the sustained provision of these ecosystem services (Adeleye et al., 2022). Recently, tropical rainforests have been identified as terrestrial systems in urgent need of global conservation efforts since they are degrading and fragmenting at alarming rates (Poorter et al., 2021). The transformation of natural rainforests into commercial lands is indeed a global phenomenon. Xishuangbanna, for instance, is a region that falls within the Indo-Burma biodiversity hotspot and boasts the largest contiguous area of tropical rainforests in China (Cao et al., 2006). More than 20 % of the rainforests

* Corresponding author.
E-mail address: lwj@xtbg.org.cn (W. Liu).

https://doi.org/10.1016/j.geoderma.2025.117197
Received 6 June 2024; Received in revised form 16 December 2024; Accepted 28 January 2025
Available online 17 February 2025

in this area have been disturbed or converted into agricultural use over the past two decades (Zhang et al., 2019). Consequently, numerous environmental issues have arisen, including soil degradation, regional water scarcity and ground water pollution (Zou et al., 2021). In particular, the hydrological balance is significantly impacted by the conversion of rainforests into monocultural plantations, such as rubber (*Hevea brasiliensis*) trees (predominantly planted in the tropics), which are kown to consume excessive amounts of water (Gnanamoorthy et al., 2022). Thus, it is imperative to take effective measures to secure and restore the rainforests with the aims to enhance their hydrological function.

Most of the rainforests on earth are located on mountain terrains, which exhibit a unique topographical setting that significantly influences the soil hydrological processes (Zhu, 1992). Additionally, tropical monsoon climates are characterized by highly uneven precipitation distribution, with about 80 % of total rainfall occurring during the rainy season. Consequently, many high-water-demanding species face water shortages during the dry seasons due to low streamflow, inadequate soil water storage, and insufficient rainfall replenishment (Chen et al., 2021). During the rainy season, however, extreme rainfall events often lead to substantial surface runoff, causing extensive soil erosion and nutrient leaching on slopes (Liu et al., 2018; Liu et al., 2016). Therefore, given the changing climate regimes, understanding the intricate interplay between climate, edaphic conditions, and hydrological processes is crucial for comprehending the overall functioning and sustainability of these biodiversity-rich ecosystems.

Soil water cotent is an essential factor that controls energy fluxes and influences the hydrological processe. It establishes the fundamental state for vegetation growth and plays a key role in regulating the hydrological dynamics of an ecosystem (Deng et al., 2016; Padilla and Pugnaire, 2007). A multitude of factors govern the dynamics of soil water content, prompting extensive research in diverse regions. In southwest China, the relationship between soil moisture and climate has been explored, with precipitation and temperature considered as the main factors (Deng et al., 2018). Ursino and Contarini (2006) evaluated the instability of soil water content advection and diffusioninstability for different rainfall regimes. These authors' findings suggest that enhancing the soil water storage capacity and improving the soil connectivity (to facilitate the lateral advection) could help mitigate the impacts of environmental change. At the watershed scale, the relative influence of soil and topographic properties on soil water content distribution was investigated, revealing stronger influences of edaphic and topographic properties found under wetter conditions in a Canadian prairie (Hu and Si 2014). On the Loess Plateau, afforestation projects have led to a depletion of soil water storage and a shift in spatial variability, highlighting the significant impacts of land-use and vegetation on soil water dynamics (Jia et al., 2017; Jin et al., 2011). Western et al. (2004) concluded that while soil characteristics and climate provide a general pointer to what we might expect in the patterns of soil water content, the results also exhibit subtleties specific to each location. To date, a substantial portion of research efforts has predominantly concentrated on temperate climate areas, whereas tropical rainforest zones have received comparatively limited attention. In the context of the ongoing UN Decade on Ecosystem Restoration (2021 – 2030), there is an urgent need to deepen our comprehension of soil water dynamics for preserving the hydrological functioning of rainforests, which are indispensable for maintaining global ecological balance and biodiversity.

The infiltration processes play a crucial role in determining runoff generation, aquifer recharge, and pollutant transport by partitioning rainfall at the soil-atmosphere interface (Morbidelli et al., 2018). In general, the infiltrating water movement can be categorized into two main types: lateral movements (or lateral flow) and vertical downward movements. The vertical movement of water flow can be further subdivided into uniform flow, which occurrs through the unsaturated zone, and preferential flow, which follows more distinct and often faster pathways through the soil (Sukhija et al., 2003). Preferential flow has been extensively studied for its intricate nature and ability to expedite groundwater recharge by swiftly conveying water into the soil profile. However, it also poses challenges as it can restrict water availability for plant uptake and potentially facilitate the transport of surface contaminants into receiving water bodies (Bargués Tobella et al., 2014). Thus, understanding the mechanisms and implications of infiltration patterns is crucial for managing regional water resources. The dye-tracer technique is a widely employed method for visualizing infiltration patterns in soil, due to its recognized reliability and high spatial resolution. This technique allows researchers to investigate the interaction between soil morphology and water movement, providing valuable insights into how water flows through the soil and where it accumulates (Allaire et al., 2009). Despite the well-established understanding that precipitation significantly affects infiltration, this factor has been overlooked in previous research designs (Jackson et al., 2005). For example, numerous dye-tracer experiments have been conducted across various land-use types, including farmlands, commercial plantations, agroforestry systems, etc (Chen et al., 2021; Zhu et al., 2019; Jiang et al., 2017, 2018). However, a notable limitation of these studies is that they only simulated a single precipitation level. Furthermore, the complexity of infiltration processes in natural conditions is exacerbated by sloping surfaces. At different slope gradients, factors such as structural cracks, faunal activities, root networks, and pedological characteristics can significantly alter pathways for water movement (Angulo-Jaramillo et al., 2016). Given that most tropical landscapes feature sloping topography and are heavily influenced by tropical monsoon climate, it is necessary to investigate how infiltration patterns will vary under different slope gradients and precipitation levels in the rainforests. We hypothesize that as precipitation increases, the infiltrated area generally expands, with soil water flow exhibiting distinct patterns: uniform flow dominate at flat sites, preferential flow dominate at moderate-slope sites, and lateral flow dominate at steep-slope sites, respectively.

In this study, soil hydrological processes, encompassing soil water content spatiotemporal dynamics and infiltration patterns were investigated, at different slopes ranging from 0° to 30° in tropical rainforests. The primary objectives were as follows: (i) to quantify the soil water storage capacity and assess the stability of soil water content at different spatiotemporal scales, (ii) to unveil the patterns of soil water flow during infiltration processes, including lateral flow, uniform flow and preferential flow, under varying precipitation levels, and (iii) to identify the relevant factors influencing soil water content and infiltration patterns within this unique ecosystem. By gaining insights into these hydrological processes, this work aims to offer constructive guidance for selecting suitable locations for afforestation projects and developing sustainable forest management strategies.

2. Material and methods

2.1. Site description and meteorological features

In this study, all field experiments were conducted at in May 2020, and soil water content was measured from November 2019 to October 2020. The study area is located in Xishuangbanna, Yunnan Province, Southwestern China (21°10′ – 22°40′N, 99°55′ – 101°50′E, Fig. S1). This region has mountainous terrain, with approximately 90 % of its area covered by hillslopes exceeding 10° in gradient, and certain areas featuring hillslopes with gradients greater than 30°. Thus, three tropical rainforest sites were chosen for this study, including a flat (0°, SF), a moderate-slope (10° north-facing, SM) and a steep-slope site (30° north-facing, SS). The vegetation in the rainforests is predominantly composed of species such as *Pometia pinnata* Forst., *Pittosporopsis kerrii* Craib, *Baccaurea ramiflora* Lour., *Barringtonia fusicarpa* Hu, *Millettia leptobotrya* Dunn, which are representative of tropical seasonal rainforests. Regarding soil type, the area is generally characterized as clay-textured laterites, also known as Oxisols.

Due to the onset of tropical southern monsoon originating from the

Table 1
Soil bulk density and textures were measured in May 2020 at 0 – 50 cm depths at three study sites.

Variable	Site	Soil depth (cm)			
		0 – 10	10 – 20	20 – 30	30 – 50
Bulk density (g cm^{-3})	SF	1.33 ± 0.02 a	1.45 ± 0.03 a	1.42 ± 0.02 a	1.41 ± 0.03 a
	SM	0.99 ± 0.05c	1.11 ± 0.05c	1.15 ± 0.02b	1.29 ± 0.04 a
	SS	1.20 ± 0.03b	1.28 ± 0.03b	1.39 ± 0.04 a	1.36 ± 0.06 a
Sand (%)	SF	57.95 ± 1.62 a	53.67 ± 1.45 a	48.33 ± 2.17 a	44.93 ± 2.01 a
	SM	44.82 ± 1.41b	43.15 ± 1.66b	42.75 ± 0.29 a	32.85 ± 0.80b
	SS	44.47 ± 2.40b	42.98 ± 0.99b	43.47 ± 3.36 a	36.14 ± 0.52b
Silt (%)	SF	12.97 ± 1.35c	13.12 ± 1.73c	14.09 ± 0.41c	20.26 ± 1.16b
	SM	19.95 ± 0.93b	18.67 ± 0.21b	17.72 ± 0.84 ab	24.59 ± 0.46 a
	SS	26.15 ± 1.14 a	26.89 ± 1.46 a	24.54 ± 4.33 a	26.34 ± 1.06 a
Clay (%)	SF	29.09 ± 2.69 a	33.21 ± 3.06 a	37.58 ± 2.55 ab	34.81 ± 1.26b
	SM	35.22 ± 0.48 a	38.18 ± 1.87 a	39.53 ± 1.01 a	42.56 ± 0.76 a
	SS	29.37 ± 2.94 a	30.13 ± 2.04 a	31.99 ± 1.99c	37.52 ± 0.84b

Data are shown as mean ± se (n = 4). Different letters indicate significant differences between sites ($p < 0.05$). SF: flat site; SM: moderate-slope site; SS: steep-slope site.

equatorial Indian Ocean (Tan et al., 2010), this region experiences a distinct dry season characterized by frequent heavy fog (November – April) and a rainy season when approximately 80 % of the total annual precipitation occurs (May – October). Between 2011 and 2020, there were significant variations in precipitation and air temperature between these two seasons (Fig. S2). Over this 10-year record period, the average annual air temperature was 23 °C, and the average annual precipitation amounted to 1230 mm. Specifically, during this time frame, 26 % of the days had precipitation exceeding 10 mm accounted for, 8 % had precipitation above 25 mm, and 1 % had precipitation above 50 mm. Notably, a decrease in dry-season precipitation was observed, suggesting a tendency towards seasonal droughts.

2.2. Soi measurements and dye-tracer experiments

2.2.1. Soil water content

Soil water content measurements were conducted at each study site over one year, with 12 measurement occasions spaced at monthly intervals. At the end of each month, three randomly distributed soil samples (about 200 g) were collected using a soil auger (diameter: 5 cm), each sample was a mix of three subsamples. The samples were taken from various depths: 0 – 10 cm, 10 – 20 cm, 20 – 30 cm, 30 – 50 cm, 50 – 70 cm, 70 – 90 cm, and 90 – 110 cm. Then, the collected soil samples were oven-dried at 105 ℃ for 48 h to determine the mass soil water content (%).

2.2.2. Soil physical properties and soil texture

The cutting ring method was employed to measure the soil physical properties. Firstly, the empty cutting rings (diameter: 70 mm, hight: 52 mm, volume: 200 cm^3) were weighed as W_{ESC} (g) and were used to cut four replicated soil cores in the field at depths of 0 – 10 cm, 10 – 20 cm, 20 – 30 cm, and 30 – 50 cm after removing the litter horizon. The cutting rings with fresh soils were weighed as W_{SCF} (g), and saturated by placing them in a stainless steel plate with distilled water. During this process, the soil samples were first placed in the container. Afterwards, the distilled water was slowly poured into the container until the water level nearly reached the height of the cutting rings, allowing it to fill the soil pores and displace any trapped air. After 20 h, the saturated samples were weighed as W_{SAT} (g) and then placed on a dry sand layer. After gravitational draining for 2 h and 5 d, samples were weighed as W_{2h} (g) and W_{5d} (g), respectively. Finally, the samples were oven-dried at 105 °C for 72 h and weighed as W_{SCD} (g). According to Chen (2005), the relevant variables were calculated using equations (1) to (7). It is worth noting that the calculation of soil porosities was based on the assumption that no air remained in the pores after saturation.

$$Bulkdensity = \frac{W_{SCD} - W_{ESC}}{200} \tag{1}$$

Where bulk density (g cm^{-3}) is the mass of dry soil per unit volume, representing the soil compaction level.

$$Saturatedwatercapacity = \frac{W_{SAT} - W_{SCD}}{W_{SCD} - W_{ESC}} \times 100\% \tag{2}$$

Where saturated water capacity (%) encompasses the maximum amount of water that soil can hold after it has been fully saturated, representing the upper limit of the soil's water-holding capacity.

$$Capillaryholdingcapacity = \frac{W_{2h} - W_{SCD}}{W_{SCD} - W_{ESC}} \times 100\% \tag{3}$$

Where capillary holding capacity (%) characteristically refers to the ability of soil to hold water against the force of gravity in its capillary spaces, which are the small spaces between soil particles.

$$Fieldcapacity = \frac{W_{5d} - W_{SCD}}{W_{SCD} - W_{ESC}} \times 100\% \tag{4}$$

Where field capacity (%) refers to the amount of water that soil can hold against the force of gravity after it has been saturated with water and excess water has drained away. This variable represents the maximum amount of water that can be held in the soil against the force of gravity.

$$Capillaryporosity = \frac{Bulkdensity \times Capillaryholdingcapacity}{\rho_{water}} \tag{5}$$

Where capillary porosity (%) refers to the portion of the total pore space in soil that is occupied by water held by capillary forces.

$$Noncapillaryporosity = \frac{Bulkdensity \times (Saturatedwatercapacity - Capillaryholdingcapacity)}{\rho_{water}} \tag{6}$$

Where noncapillary porosity (%) refers to the portion of the total pore space in soil that is not occupied by water held by capillary forces. The noncapillary pores are larger pores in the soil that are not small enough to hold water against the force of gravity, and water can freely move through them.

$$Totalporosity = Capillaryporosity + Noncapillaryporosity \tag{7}$$

Where total porosity (%) is the proportion of a soil's volume that is not occupied by soil particles and is available for air and water movement and storage, encompassing both capillary and noncapillary pores.

Soil texture was measured using the pipette method after soils were dispersed with $(NaPO_3)_6$ (van Reeuwijk, 2002). The particle compositions were examined according to the international standard for soil texture classification (Atterberg, 1911), and divided into sand (0.02 – 2 mm), silt (0.002 – 0.02 mm), and clay (< 0.002 mm).

2.2.3. Soil organic matter and root biomass

After removing the litter from the ground surface, four replicated undisturbed soil samples were collected horizon at 0 – 10 cm, 10 – 20 cm, 20 – 30 cm, and 30 – 50 cm depths. Each soil sample (about 2 kg) was taken as a composite of 4 subsamples collected at the four corners of

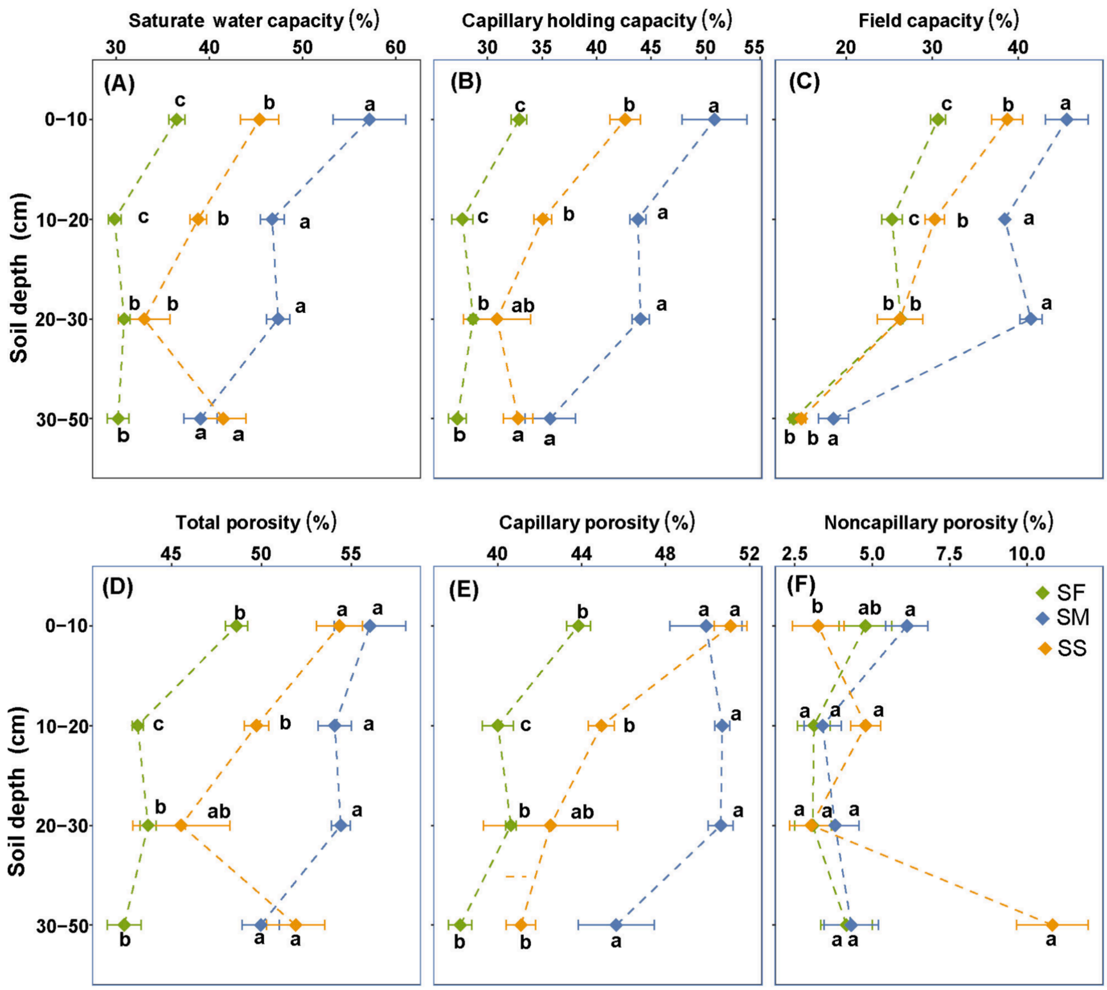

Fig. 1. Soil physical properties measured in May 2020 at 0 – 50 cm depths at three study sites. Different letters indicate significant differences between sites ($p < 0.05$), and error bars indicate the standard error of the means (n = 4). SF: flat site; SM: moderate-slope site; SS: steep-slope site.

a 5 m × 5 m plot. In the laboratory, roots, stones, and litter debris were carefully removed from the soil after being air-dried for two weeks. All soils were passed through 2 mm and 100 μm diameter sieves. Soil organic carbon concentrations were determined using the potassium dichromate oxidation-external heating method (Chapman and Pratt, 1961). The soil organic matter content was calculated by multiplying soil organic carbon values by 1.724.

Soil monoliths (10 cm × 10 cm × 10 cm) were cut with sixteen replicates at depths of 0 – 10 cm, 10 – 20 cm, 20 – 30 cm, and 30 – 50 cm. All roots were picked out and separated into fine roots (diameter < 2 mm) and coarse roots (diameter ≥ 2 mm), then oven-dried at 65 ℃ for 72 h. The dry biomass was weighed to calculate coarse, fine, and total root biomass (kg m^{-3}).

2.2.4. Dye-tracer experiments

Dye-tracer experiments were carried out in the field to monitor and analyze the infiltration patterns at each site. Five infiltration levels were set (10 mm, 25 mm, 50 mm,75 mm, and 100 mm, respectively) by referring to the precipitation records. Five steel frames (20 cm × 20 cm) were slowly inserted 5 cm into the soil after removing litter present on the ground. Brilliant Blue FCF dye solution was prepared at 4 g/L. Thereafter, the dye solution of 0.4 L, 1 L, 2 L, 3 L, and 4L was slowly poured into the frame, representing the 5 simulated infiltration levels, respectively. The frames were gently removed 24 h after the dye solution had fully infiltrated the soil. Vertical soil profiles were neatly excavated at distances of 2 cm, 5 cm, and 10 cm away from the frames' edge to expose all dyed areas for observation. Dyed profiles were photographed using a camera, with calibrated steel tape placed along the horizontal and vertical axes of the profile for reference (Fig. S3). Images were processed with Photoshop CS6 software and Adobe Illustrators CS6 software, and the relevant variables were then calculated from the output data for each image using equations (8) to (13) (Bargués Tobella et al., 2014):

$$DC = 100 \times \frac{D}{D + ND} \tag{8}$$

Where DC (%) is the total dye coverage in the vertical profile, D is the stained area (cm^2), and ND is the non-stained area (cm^2).

$$LC = 100 \times \left(\frac{D - 80{\cdot}20}{D + ND}\right) \tag{9}$$

$$LatF = 100 \times \frac{LC}{DC} \tag{10}$$

Where *LatF* (%) is the fraction of lateral flow to the total dye coverage, *LC* (%) is the coverage of lateral flow in the whole vertical profile.

$$UniF = 100 \times \frac{UID{\cdot}20}{D} \tag{11}$$

Where *UniF* (%) is the fraction of uniform flow in the total dye coverage. *UID* (cm) is the depth at which dye coverage decreases below 80 % within 20 cm width.

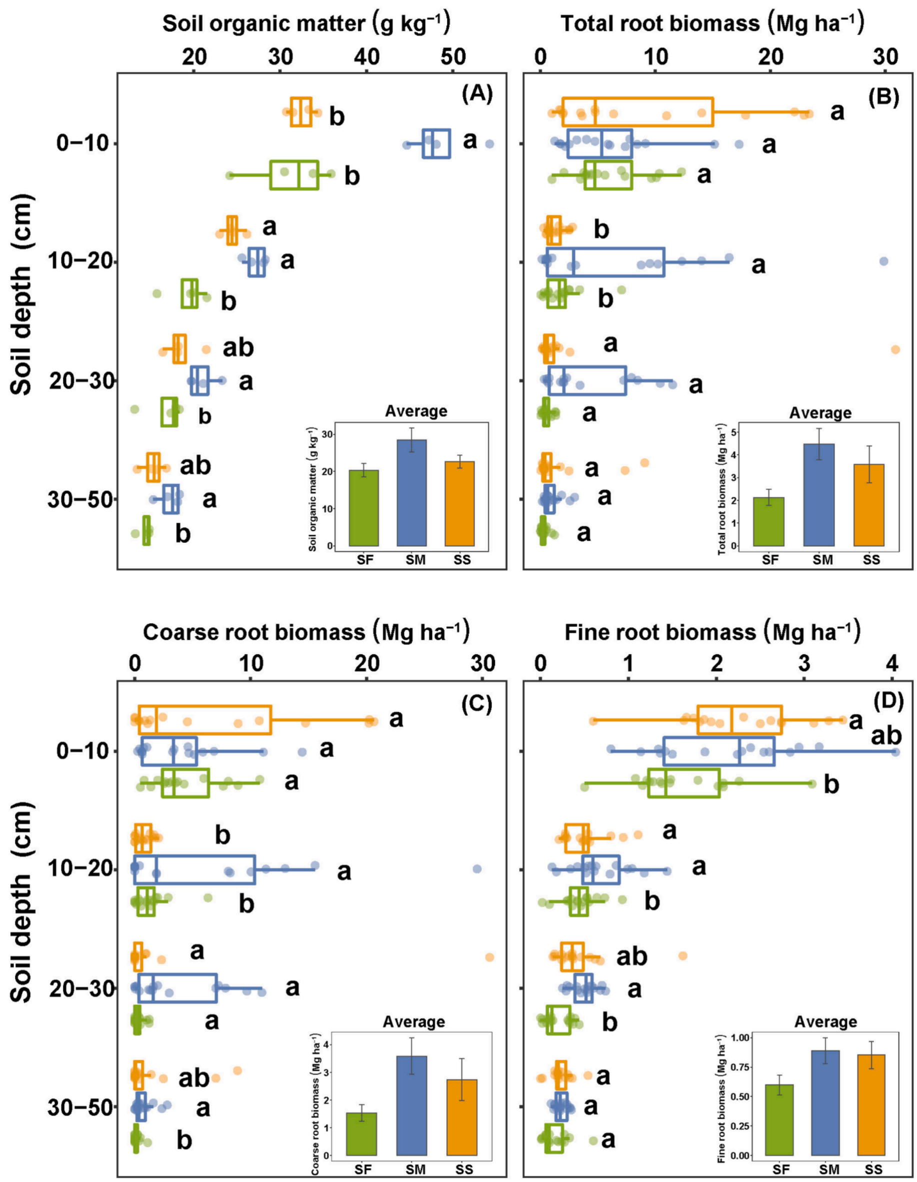

Fig. 2. Soil organic matter and root biomass measured in May 2020 at 0 – 50 cm depths at three study sites. Different letters indicate significant differences between sites ($p < 0.05$), and error bars indicate the standard error of the means (n = 4 for soil organic matter, n = 16 for root biomass). SF: flat site; SM: moderate-slope site; SS: steep-slope site.

$$PRE = 100 \times (1 - \frac{80 \cdot 20 - UID \cdot 20}{D + SD}) \tag{12}$$

$$PreF = 100 \times \frac{PRE}{DC} \tag{13}$$

Where *PreF* (%) is the fraction of preferential flow to the total dye coverage. This phenomenon enables water and solutes to infiltrate to greater depths and faster rates than uniform flow in vertical movements. PRE (%) covers preferential flow in the whole vertical profile.

2.3. Statistical analysis

All statistical analyses were performed in R software (v.4.2.2). The differences in environmental factors across study sites and the differences in infiltration variables under a specific precipitation level at each site were examined by one-way ANOVA (Tukey's post-hoc test for the pairwise comparisons of means). The temporal and spatial stability of soil water content was calculated based on the coefficient of variation (mean/sd) of soil water content across time and depth (space), respectively. The linear-effects mixed model analysis (LMM) was conducted using the 'nlme' R package (some values were log-transformed to improve the normality of the residuals). Slope gradient, air temperature, and precipitation were considered as fixed factors, while soil depth was considered as a random factor to test the effects of slope, climate and their interactions on soil water content. Slope gradient and simulated precipitation level were chosen as fixed factors, while excavated profile was considered as random factor to examine the effects of slope, precipitation and their interactions on infiltration variables. The principal component analysis (PCA) is performed by the R package of 'FactoMineR' and 'ExPosition'. Pearson's parametric regression analysis was performed using the 'lm' function. Graphs were visualized using the R packages, including 'ggplot2', 'ggpubr' and 'pheatmap'.

3. Results

3.1. Soil physical properties, soil organic matters and root biomass

The soils at three rainforest sites were similarly classified as loamy

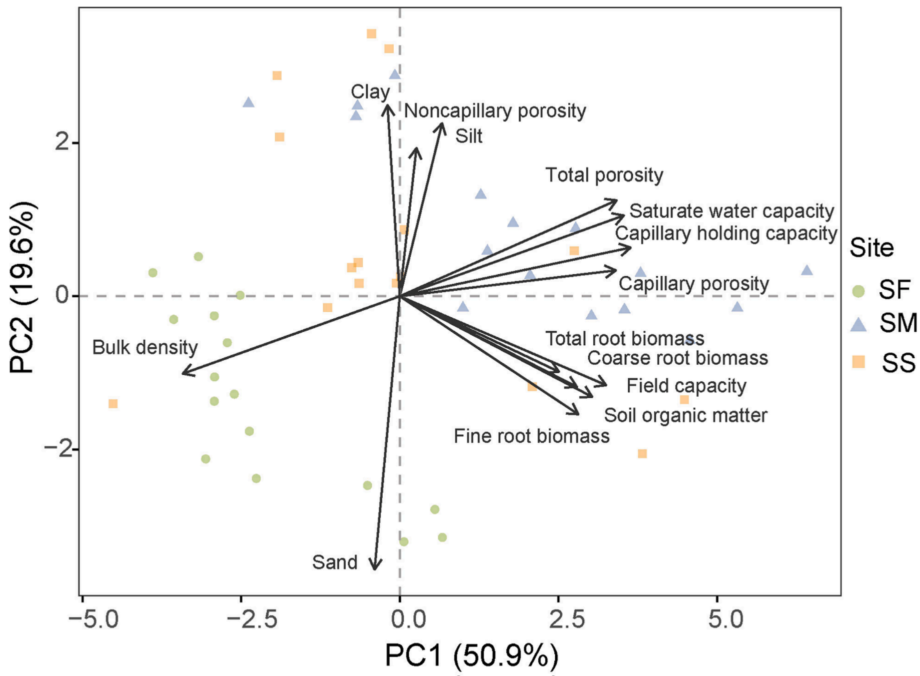

Fig. 3. Principal component analysis of the factors influencing site characteristics across three study sites (n = 16). SF: flat site; SM: moderate-slope site; SS: steep-slope site.

clay texture, with an average of 45 % sand, 20 % silt, and 35 % clay content (Table 1). Notably, slightly higher clay content was observed at the moderate-slope site than at the other two sites. Soil bulk density values at the 0 – 50 cm depths followed the order of flat site > steep-slope site > moderate-slope site, with significant differences observed within the 0 – 30 cm soil layers ($p < 0.05$). Soil physical properties (i.e., saturated water capacity, capillary holding capacity, field capacity, total porosity, capillary porosity, and noncapillary porosity) generally decreased with increasing soil depth and followed the order of moderate-slope site ≥ steep-slope site ≥ flat site (Fig. 1, $p < 0.05$). Across the 0 – 50 cm soil layers, the content of soil organic matter was highest at the moderate-slope site but lowest at the flat site (Fig. 2A). Substantial variations in root biomass were observed, not only between different study sites but also within each site, which were mainly attributed to the uneven distribution of coarse roots (Fig. 2B and Fig. 2C). Furthermore, significantly lower fine root biomass was found at the flat site compared to the sloping sites within the 10 – 30 cm soil depths (Fig. 2D, $p < 0.05$). The PCA results suggested that the moderate-slope site exhibited a relatively suitable soil condition for infiltration in comparison to the steep-slope and flat sites, which was supported by lower soil bulk density, better soil physical properties, and higher root biomass (Fig. 3). In addition, soil organic matter, root biomass and field capacity exhibited significant correlation with each other, while non-capillary capacity had significant correlation with soil texture. Moreover, drawing upon the PCA results, the key factors significantly influenced the variations in site characteristics were selected for further in-depth analysis. Specifically, capillary holding capacity, bulk density, sand content, soil organic matter, and fine root biomass were representative factors for this purpose.

3.2. Soil water spatiotemporal variations

In the rainforests, soil water content exhibited significant variations between the dry and rainy seasons, demonstrating strong responsiveness to regional climate fluctuations (Fig. 4B). Throughout the monitoring period, soil water contents along the 0 – 110 cm depth generally followed the order of moderate-slope site > steep-slope site > flat site. During the dry season, the surface layers (0 – 20 cm) were relatively drier compared to the deeper layers. Conversely, during the rainy season, an opposite trend was observed, with higher soil water contents in the surface layers than in the deeper layers (Fig. 4B). The moderate-slope site exhibited higher temporal stability in soil water content across all depths than both the flat and steep-slope sites (Fig. 4C). However, in terms of spatial stability (variation across soil depth), the flat site demonstrated the highest degree, followed by the moderate-slope and steep-slope sites (Fig. 4D). The results of the linear mixed-effects model revealed that air temperature, precipitation, and slope gradient had significant impacts on soil water content, with the most prominent effects observed in the middle (30 – 70 cm) soil layers (Table 2, $p < 0.05$). Additionally, significant interaction effects of slope, air temperature, and precipitation on soil water content were observed in the surface soil layers (0 – 20 cm). In the rainforests, soil physico-chemical properties, soil organic matter and root density can significantly affect soil water content (Table 3). Notably, soil capillary holding capacity, soil organic matter, and fine roots were positively correlated with soil water content. In contrast, soil bulk density and sand content were negatively correlated with soil water content. During the dry period, soil water content was significantly influenced by soil capillary holding capacity, bulk density, and sand content ($p < 0.05$). During the transition and wet periods, soil capillary holding capacity, bulk density, organic matter and fine root biomass had statistically significant effects on soil water content ($R^2 > 0.5$, $p < 0.05$).

3.3. Infiltration patterns

In the rainforests, the vertical movement emerged as the dominant infiltrating pathway across all study sites, yet notable variations were observed among different sites under certain precipitation levels (Fig. 5 and Fig. S4). The data revealed a significant increase in the DC as the simulated precipitation level raised, implying a general expansion of the infiltrated area with elevated precipitation (Fig. 6 and Fig. 7, $p < 0.01$). At both flat and steep-slope sites, LatF significantly increased with DC, while PreF significantly decreased. At the moderate-slope site, both LatF and UniF displayed a fluctuating pattern without a discernible trend, while PreF demonstrated a significant decreasing trend with increasing DC ($p < 0.05$). Precipitation level had a significant impact on all infiltration variables, whereas the slope effect was only significant for PRE and LatF (Table 4, $p < 0.05$). Moreover,

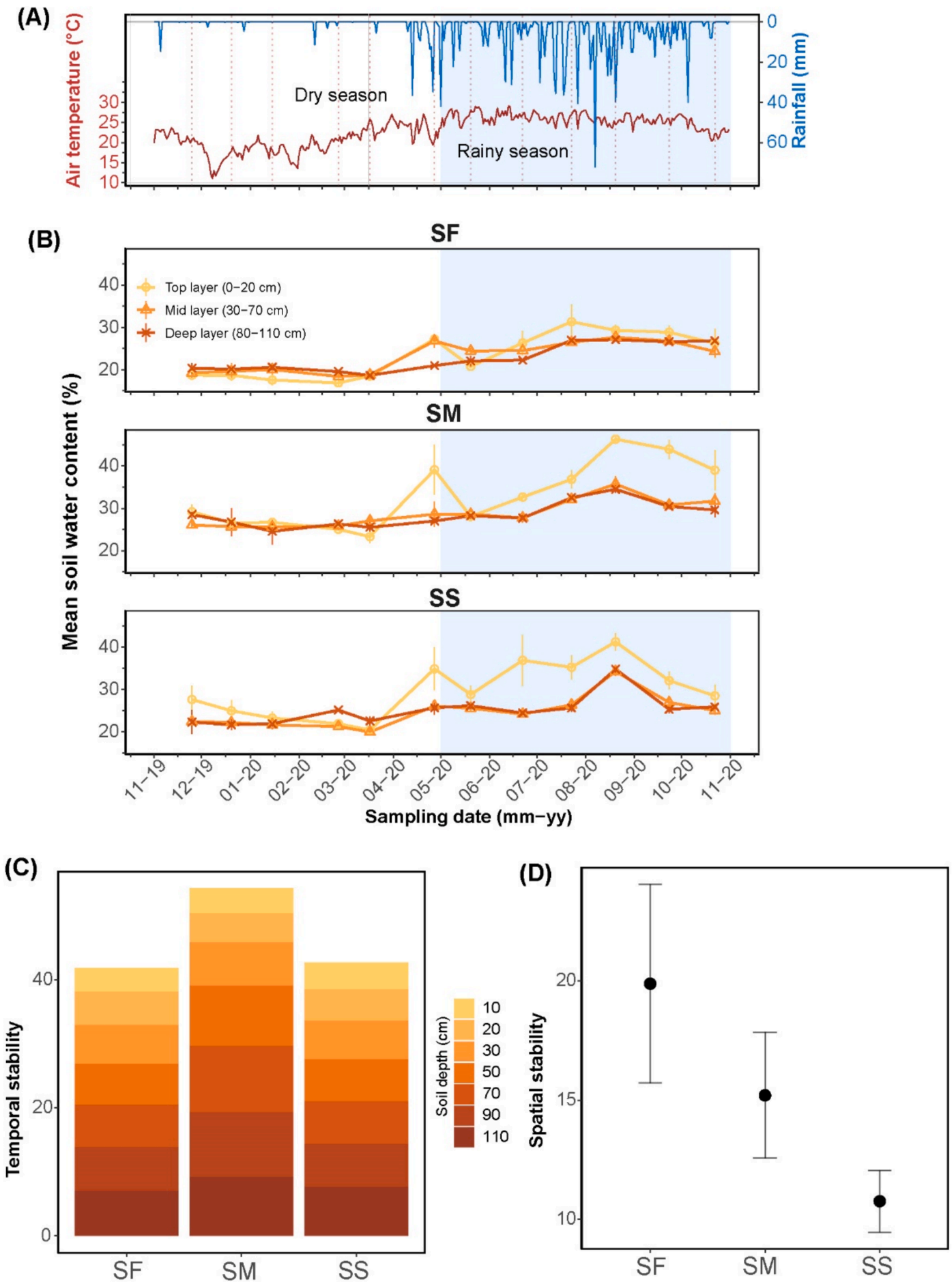

Fig. 4. Daily mean air temperature and summed up precipitation (A), and monthly measured soil water content from soil depths of 0–110 cm aggregated into three different depths – top layer, mid layer, and deep layer at three study sites (B) from November 2019 to October 2020. Temporal stability of different soil layers (C) and spatial stability (D) of soil water content across three sites. The stability is represented by the coefficient of variation of soil water content across time (temporal stability of 12 months) and depth (spatial stability of 7 depths). The error bars in (B) and (D) indicate the standard deviation. The white background shows the dry season, and the blue background shows the rainy season. SF: flat site; SM: moderate-slope site; SS: steep-slope site.

significant interactions between slope and precipitation influenced the fraction of all types of soil water flows ($p < 0.05$).

4. Discussion

4.1. Soil water stability and the controlling factors

In the present study, a strong association between soil water content and climatic factors was found across 0 – 110 cm soil layers, underscoring the importance of climatic factors in driving soil water dynamics in tropical rainforests (Fig. 4 and Table 2). The 10-year meteorological record has shown a worrying trend of increasing drought severity during the dry seasons, suggesting that rainforest functionality is becoming less resilient to ongoing climate change (Fig. S2). This vulnerability suggested that the ecological services provided by rainforests, such as water regulation and carbon sequestration, may be compromised in the near

Table 2
Linear mixed-effects models showing the effects of slope gradient (S), air temperature (T_{air}), and precipitation (P) and their interactions on soil water content at different soil depths– (top layer, mid layer and deep layer).

Factors	df	Top (0 – 20 cm)		Mid (30 – 70 cm)		Deep (80 – 110 cm)	
		F	*p*	*F*	*p*	*F*	*p*
S	11	**148.61**	**< 0.0001**	**306.75**	**< 0.0001**	**153.20**	**< 0.0001**
T_{air}	11	**301.97**	**< 0.0001**	**537.36**	**< 0.0001**	**169.66**	**< 0.0001**
P	11	**157.92**	**< 0.0001**	**410.51**	**< 0.0001**	**181.01**	**< 0.0001**
S × T_{air}	22	1.43	0.242	**3.54**	**0.030**	0.11	0.900
S × P	22	0.43	0.654	**9.32**	**< 0.0001**	1.96	0.144
T_{air} × P	22	**39.79**	**< 0.0001**	**47.52**	**< 0.0001**	**5.74**	**0.018**
S × T_{air} × P	242	**3.84**	**0.023**	0.31	0.736	2.79	0.064

Degrees of freedom (df) as well as F- and p-values are given. Bold-faced values show significant effects ($p < 0.05$).

Table 3
Pearson's parametric regression analysis between soil water content (0 – 50 cm depth) and the key factors within the soil (resulted from PCA) across all study sites.

Factors	Dry period		Transition period		Wet period	
	R^2	*p*	R^2	*p*	R^2	*p*
Capillary holding capacity	**0.66**	**0.001**	**0.72**	**< 0.001**	**0.89**	**< 0.001**
Bulk density	**<u>0.62</u>**	**0.002**	**<u>0.62</u>**	**0.002**	**<u>0.85</u>**	**< 0.001**
Sand	**<u>0.48</u>**	**0.013**	<u>0.01</u>	0.451	<u>0.03</u>	0.574
Soil organic matter	0.14	0.236	**0.59**	**0.003**	**0.67**	**0.001**
Fine root biomass	0.08	0.362	**0.59**	**0.004**	**0.49**	**0.012**

R^2 is the determination coefficient; bold-faced values show significant relationships ($p < 0.05$); underlined values represent negative correlations, while non-underlined values represent positive correlations. Dry period: December–March; transition period: November, April – May 2020 and Oct 2020; wet period: June 2020 – September 2020.

future. However, most research endeavors pertaining to soil water dynamics have focused predominantly on arid and semi-arid regions (Liu and Shao, 2014; Zhao et al., 2017), with tropical regions consistently receiving scant attention. The prolonged and intensified seasonal droughts resulting from climate change have profound impacts on various tropical ecosystems, as restricted soil water availability significantly impedes vegetation productivity and ecosystem health (Chen et al., 2021). Thus, there is an urgent need to address this issue by implementing measures to enhance rainforest resilience and adaptability to climate change, such as promoting sustainable forest management practices and supporting research into rainforest conservation and restoration techniques. Safeguarding rainforests to uphold and enhance their regulatory role in hydrological processes is essential for sustaining the tropical water cycle. In order to maintain the water supply balance amidst different climatic conditions, the adoption of timely and precise irrigation practices is necessary. Furthermore, the construction of dams can also be a valuable strategy for managing water resources effectively.

The variation in soil water content among the three sites indicated differing water retention capabilities along mountain slopes in tropical rainforests (Fig. 4). Some studies suggest a negative correlation between slope gradient and soil water content, implying that steeper slopes retain less soil water, while others found no significant relationship (Gómez-Plaza et al., 2001). In the present study, a significant influence of slope gradient on the distribution of soil water content throughout 0 – 110 cm soil depth was found, albeit a non-linear relationship. Soil water content decreased from the flat site to the moderate-slope site and further to the steep-slope site, suggesting that factors such as soil properties and plant roots may interact with slope gradient to influence soil water allocation. At global scales, precipitation and air temperature characteristics strongly influence soil water content variability (Vivoni et al., 2008). However, at regional scales, additional factors such as topography, soil properties, and land use become crucial determinants of soil water content's spatial distribution (Zucco et al., 2014; Seneviratne et al., 2010). The findings of this study emphasize the impact of site characteristics on soil water dynamics (Table 3). These factors exhibited varying degrees of influence over time, resulting in asynchronous soil water content dynamics across different sites. During the dry period, soil texture and other physical properties predominantly governed soil water content. However, during the transition and wet periods, physical properties and biotic factors (i.e., fine root biomass and organic matter) jointly controlled soil water content. In the dry seasons, high air temperature and scant rainfall lead to significant soil pore water evaporation, making a favorable physical architecture crucial for soil water retention. Conversely, in the rainy seasons, frequent and heavy rainfall necessitates the soil's capacity to store ample water to support vegetation growth and a good drainage system at the same time. Soil clays, organic matter, and root exudates are known to be important in binding soil particles to form aggregates, facilitating channels and porosities (Liu et al., 2019; Guo et al., 2019; Jiang et al., 2018; Gómez-Plaza et al., 2001). The PCA results indicated that, among three study sites, the soil condition was the best at the moderate-slope site and the worst at the flat site (Fig. 3). This alignment between site characteristics and soil water content dynamics suggested that a favourable soil environment not only can improve the overall soil water content level but also maintain the soil water content stability. Nevertheless, long-term constant observation of soil water content and more holistic investigation of soil characteristics, including the physical, chemical and biological aspects, are encouraged in future research to gain a deeper understanding.

4.2. Infiltration patterns in response to precipitation change

Based on the analysis of the dye-tracer experiments, precipitation levels were found to have a substantial influence on water flow behaviours during infiltration (Table 4). Additionally, the interaction between precipitation and slope gradient resulted in significant changes in the fractions of lateral and vertical flows. These findings indicated the complex interplay between precipitation, slope, and site-specific characteristics in shaping infiltration patterns. Morbidelli et al. (2018) reviewed several experimental studies on infiltration on sloping surfaces and noted contradictory results, highlighting the intricate relationship between slope and infiltration influenced by various factors such as rainfall intensity, microtopography, vegetation, and soil property heterogeneity. Also, initial soil moisture conditions can affect soil infiltrability (Zhu et al., 2014).To minimize these effects in the present study, the dye-tracer experiments were simultaneously conducted at three study sites in May (the transition period between the dry and the rainy seasons), ensuring the soils were neither excessively dry nor overly wet. Despite similar soil water content across three sites, some fieldwork uncertainties are unavoidable.

At both flat and steep-slope sites, increased precipitation significantly enhanced lateral flow, while reducing vertical water movements since the preferential flow fractions were decreased (Fig. 7). This result indicated reduced water availability for recharging soil profile at these

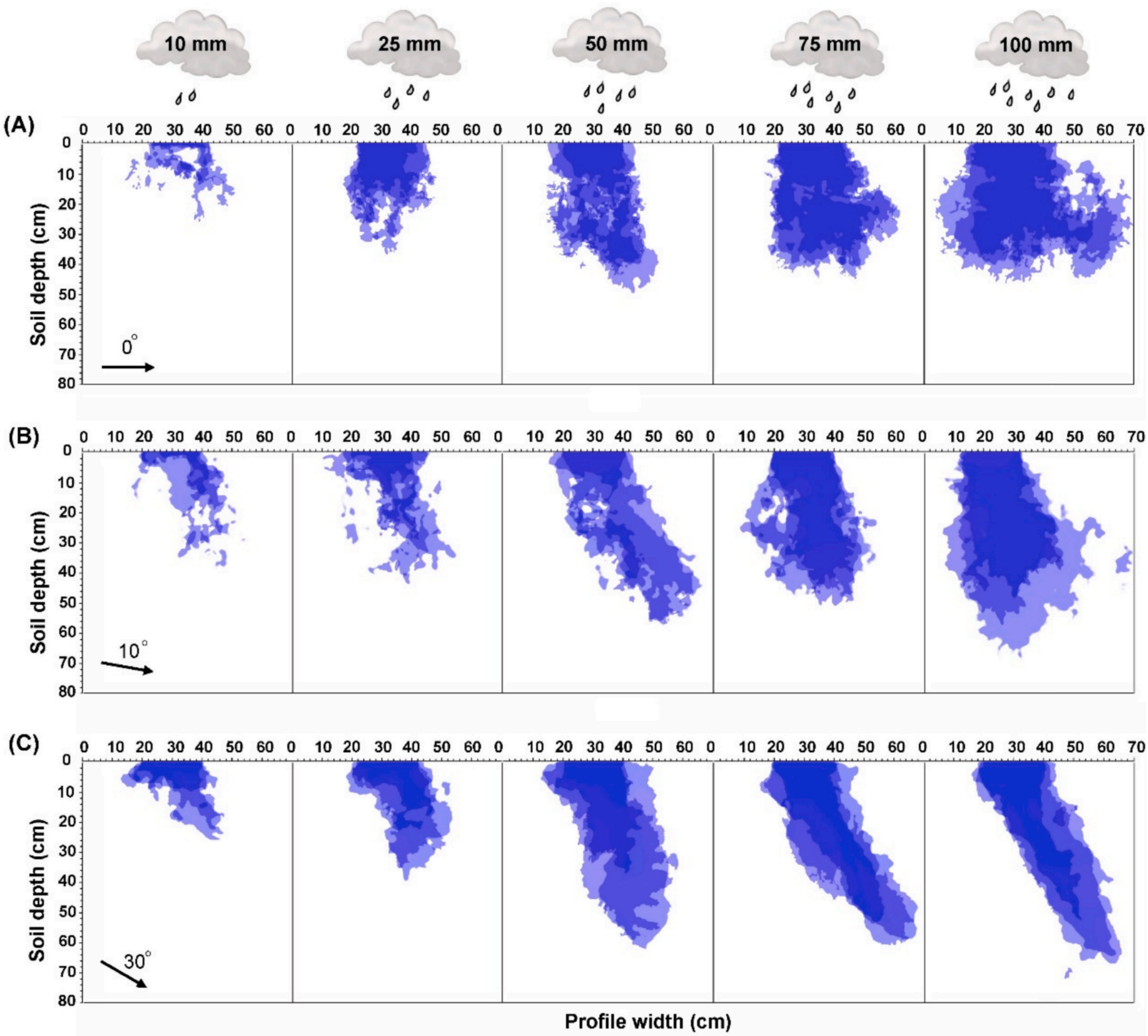

Fig. 5. Infiltration patterns visualized by dye-tracer under five simulated precipitation levels at flat (A), moderate-slope (B), and steep-slope (C) sites. Blue zone: dye-stained, white zone: non-stained. The photographs were taken 24 h after the infiltration event, and the size of the soil profiles excavated for photographs is indicated by the profile width and soil depth.

sites, particularly under extreme rainfall conditions where soils become waterlogged. This kind of lateral flow dominance at specific slope positions has also been reported in boreal forests and the loess plateau, which is attributed to variations in rock depths and soil cracks (Qi et al., 2020; Mei et al., 2018; Laine-Kaulio et al., 2015). In this work, deeper vertical water flow was observed at the sloping sites, while wider lateral water flow was dominant at the flat site. Rock fragments, root networks, faunal activities, and soil physical architectures are recognized as the main factors contributing to soil heterogeneity, facilitating preferential pathways, thereby promoting vertical water movements on hillslopes (Chen et al., 2021; Marquart et al., 2020; Zhu et al., 2019; Jiang et al., 2017). It has been reported that higher root density predominantly facilitates rapid water infiltration into deeper soil layers during extreme rainfall events (Zuo et al., 2021). Fine roots, in particular, play an important role by increasing soil organic matter, improving aggregation, and forming pore spaces (Cui et al., 2019; Wu et al., 2017). Consistent with prior studies, this study found higher levels of fine roots and soil organic matter at sloping sites than at the flat site, resulting in a significant decrease in preferential flow as the infiltrated area increased (Bargués Tobella et al., 2014). When precipitation exceeded 10 mm, a greater portion of preferential flow was observed at the moderate-slope site compared to the other two sites, indicating that moderate-slope terrain favours drainage and groundwater recharge during the rainy seasons.

4.3. Implications for forest management

In recent years, many tropical monsoon regions have witnessed a surge in the intensity and frequency of extreme climate events, posing significant challenges for biodiversity-rich forest management (Song et al., 2022). Notably, the dry seasons bring about narrow streamflow and low soil water storage (Chen et al., 2021), while the rainy seasons bring accelerated surface runoff and soil erosion (Zhu et al., 2022). Therefore, optimizing the hydrological functions of tropical rainforests is crucial for mitigating climate change. Specifically, soil water dynamics and infiltration patterns are intricately linked to regional hydraulic redistribution and nutrient transportation, significantly impacting plant water use efficiency and vegetation growth (Jiang et al.,

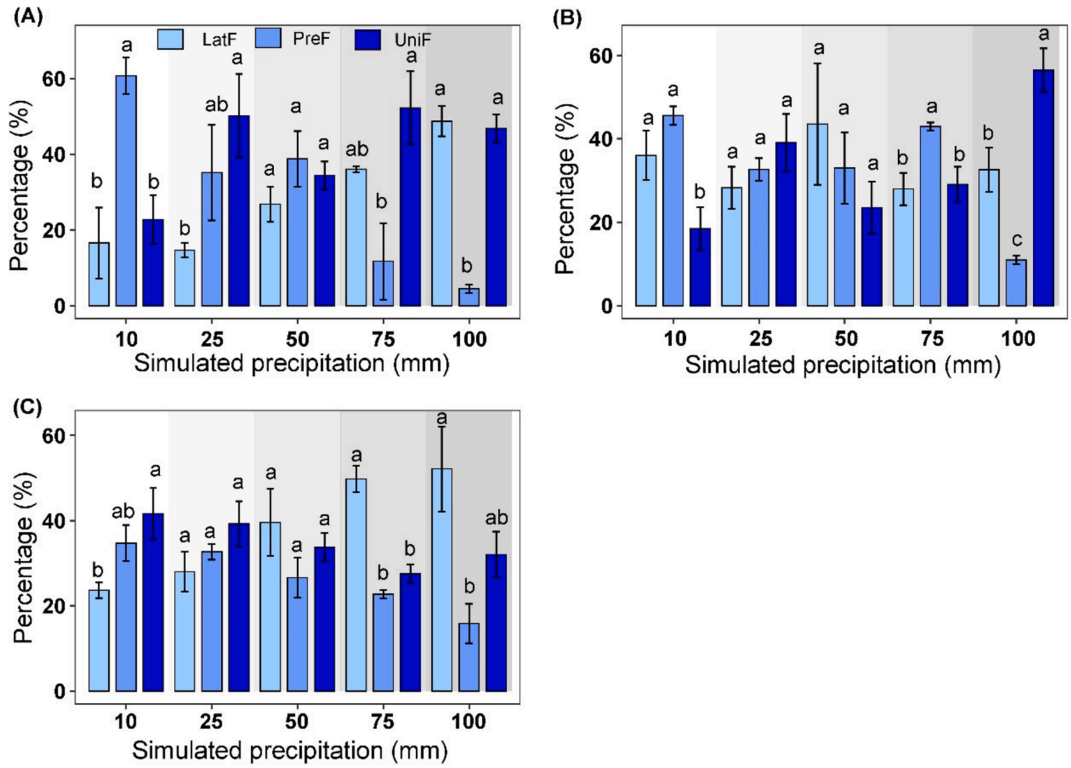

Fig. 6. The distribution of infiltrated water flow under five simulated precipitation levels at the flat (A), moderate-slope (B), and steep-slope (C) sites. Different letters indicate significant differences between water flow fractions ($p < 0.05$), and error bars indicate the standard error of the means (n = 3). UniF: uniform flow fraction, LatF: lateral flow fraction, PreF: preferential flow fraction.

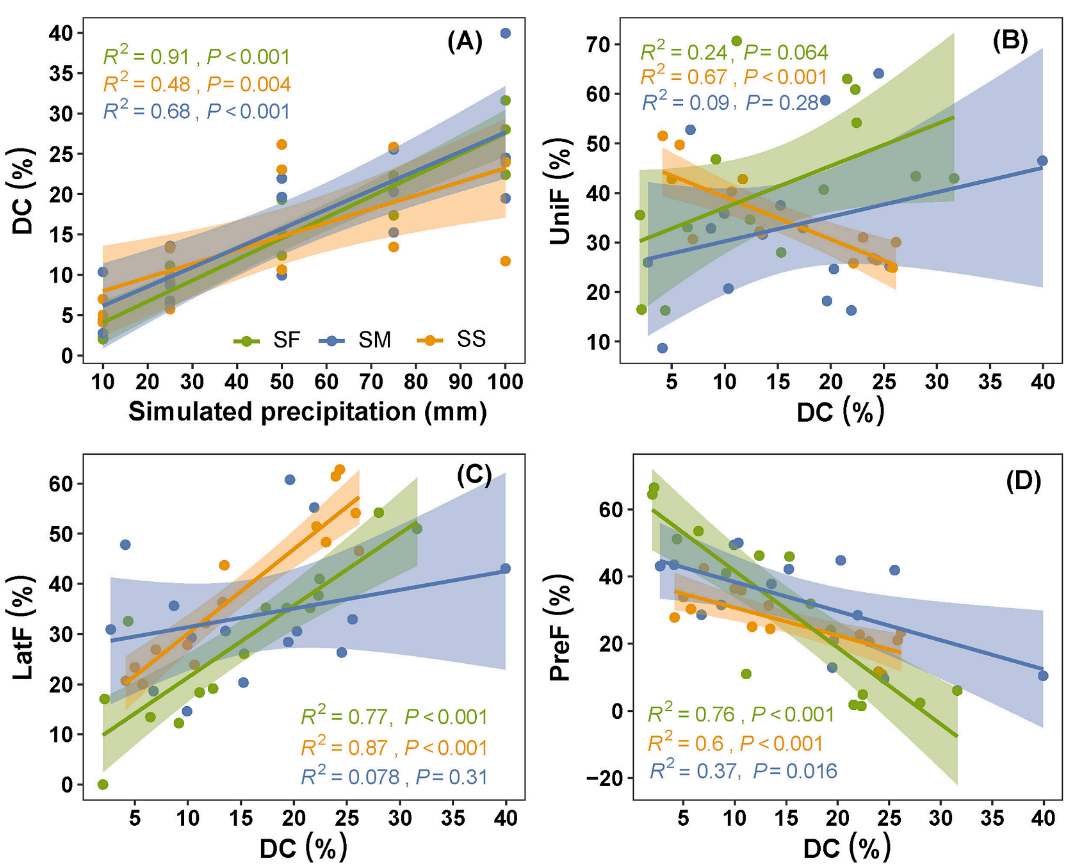

Fig. 7. Relationship between infiltration variables obtained from dye-tracer experiments at three study sites (n = 15). DC: dye coverage; UniF: uniform flow fraction; LatF: lateral flow fraction; PreF: preferential flow fraction. SF: flat site; SM: moderate-slope site; SS: steep-slope site.

Table 4
Linear mixed-effects models showing the effects of slope gradient (S) and simulated precipitation level (P) on infiltration variables.

Factors	S (df = 2)		P (df = 4)		S × P (df = 8)	
	F	*p*	*F*	*p*	*F*	*p*
DC	0.80	0.459	**82.49**	**< 0.0001**	1.89	0.102
LC	2.67	0.087	**41.58**	**< 0.0001**	1.97	0.089
PRE	**4.93**	**0.015**	**18.15**	**< 0.0001**	**3.72**	**0.005**
UID	3.02	0.07	**62.65**	**< 0.0001**	**7.58**	**< 0.0001**
UniF	1.82	0.178	**4.41**	**0.043**	**5.21**	**0.010**
PreF	1.40	0.260	**50.76**	**< 0.0001**	**5.79**	**0.007**
LatF	**5.67**	**0.007**	**37.95**	**< 0.0001**	**12.13**	**< 0.001**

Degrees of freedom (df) as well as F- and *p*-values are given. UniF: uniform flow fraction; PreF: preferential flow fraction; LatF: lateral flow fraction. Bold-faced values show significant effects ($p < 0.05$).
Note: Kindly check *Correspondence and requests for materials should be addressed to Xiaojin Jiang and Wenjie Liu.

2017). Among the three sites examined, the flat site exhibited the driest condition due to its limited water-holding capacity, low organic matter content, and poor root network (Fig. 4). Consequently, plant survival at the flat site may be threatened by water scarcity, particularly during extreme drought events in the dry season. Conversely, during extreme precipitation events in the rainy seasons, the flat site is prone to generating substantial surface runoff due to poor drainage, as evidenced by higher lateral flow but less preferential flow in dye-tracer experiments. Between the two sloping sites, the moderate-slope site stood out for its relatively high and temporally stable soil water content, suggesting that it is less vulnerable to climate change-induced drought. Thereby, the moderate-slope site can provide a more favourable and stable soil environment for plants and soil fauna. Furthermore, the constant distribution of soil water content across the 0 – 110 cm layers at the moderate-slope site can enable plants to utilize water from both surface and deep soil layers, promoting niche comlementation in soil water resource allocation. Additionally, the moderate-site's superior capacity to sustain vertical percolation during infiltration ensures that more water is retained on-site, reducing lateral flow losses. Studies have shown that trees prefere to grow more roots in soil layers with higher moisture. In turn, these well-developed root systems can help maintain soil water (Ma and Song, 2016). Therefore, we emphasize the importance of site-specific designs to optimize water resource efficiency and ensure sustainable forest management. Specifically, prioritizing the conservation of rainforests on sloping sites is highly recommended with aims to mitigate surface runoff, prevent soil erosion, and combat land degradation. In addition, forest restoration projects are suggested to be implemented on moderate slopes. Moreover, these findings implied that soil amendation is needed to improve hydrological functions, with biochar application being a potential approach (Maroušek et al., 2023). Furthermore, choosing a diverse range of species with varying rooting depths during forest restoration can avoid inter-species water competition. Hence, understanding the growth characteristics of different tree species is crucial for the success of targeted restoration efforts. Moreover, numerous other biotic factors can influence the hydrological processes within an ecosystem, such as the distinctive root morphology and water use strategies adopted by individual species. With the assistance of artificial intelligence, further research at the species level can be well designed to enhance our knowledge of the hydrological function of rainforests (Kliestik et al., 2023).

5. Conclusions

By investigating the soil water spatiotemporal dynamics and infiltration patterns at different slopes, this study emphasises the effects of climatic and edaphic factors on soil hydrological processes in mountainous rainforests. The impact of climate factors on soil water content varied across slope gradients, owing to the distinct site characteristics. Relatively higher and more stable distribution of soil water content at the moderate-slope site was observed, suggesting lower vulnerability to seasonal droughts at such location. Moreover, the moderate-slope site exhibited a better balance between lateral and vertical water movements with varied simulated precipitation levels, indicating its advanced drainage and groundwater recharge system. These better hydrological conditions at the moderate-slope site than the flat and steep-slope sites are attributed to its favourable soil environment, which characterized by good physical structure, rich soil organic matter, and dense root network. Therefore, this study underscores the nescessity of taking site-specific strategies with consideration of the hydrological functions in future rainforest stewardship. From this point of view, protecting rainforests is recommended on sloping terrains, and moderate-slope sites are strongly suggested as ideal restoration locations.

CRediT authorship contribution statement

Xin Zou: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ankit Shekhar:** Writing – review & editing, Methodology. **Yuxuan Mo:** Writing – review & editing, Visualization, Methodology. **Ashutosh Kumar Singh:** Writing – review & editing. **Xiaojin Jiang:** Writing – review & editing, Supervision. **Wenjie Liu:** Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (32371608, 32271648), the Yunnan Fundamental Research Projects (202101AS070010, 202201AT070216), and China Scholarship Council (202204910196). We thank Xishuangbanna Station of Tropical Rainforest Ecosystem Studies, Chinese Academy of Sciences and Institutional Center for Shared Technologies and Facilities of Xishuangbanna Tropical Botanical Garden, CAS for providing the required facilities. Additionally, we thank Mr. Mengnan Liu for his assistance in the field work, and Dr. Palingamoorthy Gnanamoorthy for his advice in data analysis.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geoderma.2025.117197.

Data availability

Data will be made available on request.

References

Adeleye, M.A., Connor, S.E., Herbert, A., Haberle, S.G., 2022. Retracted: Toward the saving of global rainforests. Glob. Chang. Biol. 28 (16), 5006. https://doi.org/10.1111/gcb.16112.

Allaire, S.E., Roulier, S., Cessna, A.J., 2009. Quantifying preferential flow in soils: A review of different techniques. J. Hydrol. 378 (1–2), 179–204. https://doi.org/10.1016/j.jhydrol.2009.08.013.

Angulo-Jaramillo, R., Bagarello, V., Iovino, M., Lassabatère, L., 2016. Infiltration measurements for soil hydraulic characterization. Springer International Publishing.

Atterberg, A., 1911. Die plastizität der tone. Int. Mitt. Bodenkd. 1, 4–43.

Bargués Tobella, A., Reese, H., Almaw, A., Bayala, J., Malmer, A., Laudon, H., Ilstedt, U., 2014. The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. Water Resour. Res. 50, 3342–3354. https://doi.org/10.1002/2013WR015197.

Cao, M., Zou, X., Warren, M., Zhu, H., 2006. Tropical forests of xishuangbanna. China. Biotropica 38 (3), 306–309. https://doi.org/10.1111/j.1744-7429.2006.00146.x.
Chapman, H.D., Pratt, P.F., 1961. Methods of analysis for soils, plants and water. University California, Berkeley, CA, USA.
Chen, C., Zou, X., Singh, A.K., Zhu, X., Zhang, W., Yang, B., Jiang, X., Liu, W., 2021. Effects of hillslope position on soil water infiltration and preferential flow in tropical forest in southwest China. J. Environ. Manage. 299, 113672. https://doi.org/10.1016/j.jenvman.2021.113672.
Chen, L.X., 2005. Soil experiment and practice course. Northeast Forestry University Press, Harbin.
Cui, Z., Wu, G.L., Huang, Z., Liu, Y., 2019. Fine roots determine soil infiltration potential than soil water content in semi-arid grassland soils. J. Hydrol. 578, 124023. https://doi.org/10.1016/j.jhydrol.2019.124023.
Deng, L., Yan, W., Zhang, Y., Shangguan, Z., 2016. Severe depletion of soil moisture following land-use changes for ecological restoration: Evidence from northern China. Forest Ecol. Manag. 366, 1–10. https://doi.org/10.1016/j.foreco.2016.01.026.
Deng, Y., Wang, S., Bai, X., Wu, L., Cao, Y., Li, C., Li, H., Hu, Z., 2018. Relationship between soil moisture and climate and its memory in Southwest China. Acta Ecol. Sin. 38, 8688–8699.
Gnanamoorthy, P., Song, Q., Zhao, J., Zhang, Y., Zhang, J., Lin, Y., Zhou, L., Bibi, S., Sun, C., Yu, H., Zhou, W., Sha, L., Wang, S., Chakraborty, S., Burman, P.K.D., 2022. Seasonal fog enhances crop water productivity in a tropical rubber plantation. J. Hydrol. 128016. https://doi.org/10.1016/j.jhydrol.2022.128016.
Gómez-Plaza, A., Martínez-Mena, M., Albaladejo, J., Castillo, V.M., 2001. Factors regulating spatial distribution of soil water content in small semiarid catchments. J. Hydrol. 253 (1–4), 211–226. https://doi.org/10.1016/S0022-1694(01)00483-8.
Guo, L., Liu, Y., Wu, G.L., Huang, Z., Cui, Z., Cheng, Z., Zhang, R.Q., Tian, F.P., He, H.H., 2019. Preferential water flow: Influence of alfalfa (Medicago sativa L.) decayed root channels on soil water infiltration. J. Hydrol. 578, 124019. https://doi.org/10.1016/j.jhydrol.2019.124019.
Hu, W., Si, B.C., 2014. Revealing the relative influence of soil and topographic properties on soil water content distribution at the watershed scale in two sites. J. Hydrol. 516, 107–118. https://doi.org/10.1016/j.jhydrol.2013.10.002.
Jackson, R.B., Jobbagy, E.G., Avissar, R., Roy, S.B., Barrett, D.J., Cook, C.W., Farley, K. A., le Maitre, D.C., McCarl, B.A., Murray, B.C., 2005. Trading water for carbon with biological carbon sequestration. Science 310, 1944–1947. https://doi.org/10.1126/science.1119282.
Jia, X., Shao, M., Zhu, Y., Luo, Y., 2017. Soil moisture decline due to afforestation across the Loess Plateau. China. J. Hydrol. 546, 113–122. https://doi.org/10.1016/j.jhydrol.2017.01.011.
Jiang, X.J., Liu, W., Chen, C., Liu, J., Yuan, Z.Q., Jin, B., Yu, X., 2018. Effects of three morphometric features of roots on soil water flow behavior in three sites in China. Geoderma 320, 161–171. https://doi.org/10.1016/j.geoderma.2018.01.035.
Jiang, X.J., Liu, W., Wu, J., Wang, P., Liu, C., Yuan, Z.Q., 2017. Land degradation controlled and mitigated by rubber-based agroforestry systems through optimizing soil physical conditions and water supply mechanisms: A case study in Xishuangbanna. China. Land Degrad. Dev. 28 (7), 2277–2289. https://doi.org/10.1002/ldr.2757.
Jin, T.T., Fu, B.J., Liu, G.H., Wang, Z., 2011. Hydrologic feasibility of artificial forestation in the semi-arid Loess Plateau of China. Hydrol. Earth Syst. Sci. 15, 2519–2530. https://doi.org/10.5194/hess-15-2519-2011.
Jung, M., Arnell, A., de Lamo, X., García-Rangel, S., Lewis, M., Mark, J., Merow, C., Miles, L., Ondo, I., Pironon, S., Ravilious, C., Rivers, M., Schepaschenko, D., Tallowin, O., van Soesbergen, A., Govaerts, R., Boyle, B.L., Enquist, B.J., Feng, X., Gallagher, R., Maitner, B., Meiri, S., Mulligan, M., Ofer, G., Roll, U., Hanson, J.O., Jetz, W., Marco, M.D., McGowan, J., Rinnan, D.S., Sachs, J.D., Lesiv, M., Adams, V. M., Andrew, S.C., Burger, J.R., Hannah, L., Marquet, P.A., McCarthy, J.K., Morueta-Holme, N., Newman, E.A., Park, D.S., Roehrdanz, P.R., Svenning, J.C., Violle, C., Wieringa, J.J., Wynne, G., Fritz, S., Strassburg, B.B.N., Obersteiner, M., Kapos, V., Burgess, N., Schmidt-Traub, G., Visconti, P., 2021. Areas of global importance for conserving terrestrial biodiversity, carbon and water. Nat. Ecol. Evol. 5, 1499–1509. https://doi.org/10.1038/s41559-021-01528-7.
Kliestik, T., Nica, E., Durana, P., Popescu, G.H., 2023. Artificial intelligence-based predictive maintenance, time-sensitive networking, and big data-driven algorithmic decision-making in the economics of Industrial Internet of Things. Oecon. Copernic. 14 (4), 1097–1138. https://doi.org/10.24136/oc.2023.033.
Laine-Kaulio, H., Backnäs, S., Koivusalo, H., Laurén, A., 2015. Dye tracer visualization of flow patterns and pathways in glacial sandy till at a boreal forest hillslope. Geoderma 259, 23–34. https://doi.org/10.1016/j.geoderma.2015.05.004.
Liu, B.X., Shao, M., 2014. Estimation of soil water storage using temporal stability in four land uses over 10 years on the Loess Plateau. China. J. Hydrol. 517, 974–998. https://doi.org/10.1016/j.earscirev.2019.102962.
Liu, J., Liu, W., Zhu, K., 2018. Throughfall kinetic energy and its spatial characteristics under rubber-based agroforestry systems. Catena 161, 113–121. https://doi.org/10.1016/j.catena.2017.10.014.
Liu, W., Zhu, C., Wu, J., Chen, C., 2016. Are rubber-based agroforestry systems effective in controlling rain splash erosion? Catena 147, 16–24. https://doi.org/10.1016/j.catena.2016.06.034.
Liu, Y., Cui, Z., Huang, Z., López-Vicente, M., Wu, G.L., 2019. Influence of soil moisture and plant roots on the soil infiltration capacity at different stages in arid grasslands of China. Catena 182, 104147. https://doi.org/10.1016/j.catena.2019.104147.
Ma, Y., Song, X., 2016. Using stable isotopes to determine seasonal variations in water uptake of summer maize under different fertilization treatments. Sci. Total Environ. 550, 471–483. https://doi.org/10.1016/j.scitotenv.2016.01.148.
Maroušek, J., Minofar, B., Maroušková, A., Strunecký, O., Gavurová, B., 2023. Environmental and economic advantages of production and application of digestate biochar. Environ. Technol. Innov. 30, 103109. https://doi.org/10.1016/j.eti.2023.103109.
Marquart, A., Eldridge, D.J., Geissler, K., Lobas, C., Blaum, N., 2020. Interconnected effects of shrubs, invertebrate-derived macropores and soil texture on water infiltration in a semi-arid savanna rangeland. Land Degrad. Dev. 31 (16), 2307–2318. https://doi.org/10.1002/ldr.3598.
Mei, X., Zhu, Q., Ma, L., Zhang, D., Wang, Y., Hao, W., 2018. Effect of stand origin and slope position on infiltration pattern and preferential flow on a Loess hillslope. Land Degrad. Dev. 29 (5), 1353–1365. https://doi.org/10.1002/ldr.2928.
Morbidelli, R., Saltalippi, C., Flammini, A., Govindaraju, R.S., 2018. Role of slope on infiltration: A review. J. Hydrol. 557, 878–886. https://doi.org/10.1016/j.jhydrol.2018.01.019.
Padilla, F.M., Pugnaire, F.I., 2007. Rooting depth and soil moisture control Mediterranean woody seedling survival during drought. Funct. Ecol. 21, 489–495. https://www.jstor.org/stable/4540047.
Poorter, L., Craven, D., Jakovac, C.C., Van Der Sande, M.T., Amissah, L., Bongers, F., Chazdon, R.L., Farrior, C.E., Kambach, S., Meave, J.A., Muñoz, R., Norden, N., Rüger, N., Van Breugel, M., Almeyda Zambrano, A.M., Amani, B., Andrade, J.L., Brancalion, P.H.S., Broadbent, E.N., De Foresta, H., Dent, D.H., Derroire, G., Dewalt, S.J., Dupuy, J.M., Durán, S.M., Fantini, A.C., Finegan, B., Hernández-Jaramillo, A., Hernández-Stefanoni, J.L., Hietz, P., et al., 2021. Multidimensional tropical forest recovery. Science 374 (6573), 1370–1376. https://www.science.org/doi/abs/10.1126/science.abh3629.
Qi, W., Zhang, Z., Wang, C., Chen, Y., Zhang, Z., 2020. Crack closure and flow regimes in cracked clay loam subjected to different irrigation methods. Geoderma 358, 113978. https://doi.org/10.1016/j.geoderma.2019.113978.
Seneviratne, S.I., Corti, T., Davin, E.L., Hirschi, M., Jaeger, E.B., Lehner, I., Orlowsky, B., Teuling, A.J., 2010. Investigating soil moisture-climate interactions in a changing climate: a review. Earth-Sci. Rev. 99 (3–4), 125–161. https://doi.org/10.1016/j.earscirev.2010.02.004.
Song, L., Yang, B., Liu, L., Mo, Y., Liu, W., Meng, X., Lu, H., Li, Y., Zakari, S., Tan, Z., Fan, Z., Zhang, Y., 2022. Spatial-temporal differentiations in water use of coexisting trees from a subtropical evergreen broadleaved forest in Southwest China. Agric. for. Meteorol. 316, 108862. https://doi.org/10.1016/j.agrformet.2022.108862.
Sukhija, B.S., Reddy, D.V., Nagabhushanam, P., Hussain, S., 2003. Recharge processes: Piston flow vs preferential flow in semi-arid aquifers of India. Hydrogeol. J. 11, 387–395. https://doi.org/10.1007/s10040-002-0243-3.
Tan, Z., Zhang, Y., Yu, G., Sha, L., Tang, J., Deng, X., Song, Q., 2010. Carbon balance of a primary tropical seasonal rain forest. J. Geophys. Res. Atmos. 115 (D4). https://doi.org/10.1029/2009JD012913.
Ursino, N., Contarini, S., 2006. Stability of banded vegetation patterns under seasonal rainfall and limited soil moisture storage capacity. Adv. Water Resour. 29, 1556–1564. https://doi.org/10.1016/j.advwatres.2005.11.006.
van Reeuwijk, L.P. Procedures for Soil Analysis (sixth edition). 2002. International Soil Reference and Information Centre (ISRIC), Wageningen, The Netherlands.
Vivoni, E. R., Gebremichael, M., Watts, C. J., Bindlish, R., Jackson, T. J., 2008. Comparison of ground-based and remotely-sensed surface soil moisture estimates over complex terrain during SMEX04. Remote Sens. Environ. 2008, 112(2), 314–325. https://doi.org/10.1016/j.rse.2006.10.028.
Western, A.W., Zhou, S.L., Grayson, R.B., McMahon, T.A., Blöschl, G., Wilson, D.J., 2004. Spatial correlation of soil moisture in small catchments and its relationship to dominant spatial hydrological processes. J. Hydrol. 286 (1–4), 113–134. https://doi.org/10.1016/j.jhydrol.2003.09.014.
Wu, G.L., Liu, Y., Yang, Z., Cui, Z., Deng, L., Chang, X.F., Shi, Z.H., 2017. Root channels to indicate the increase in soil matrix water infiltration capacity of arid reclaimed mine soils. J. Hydrol. 546, 133–139. https://doi.org/10.1016/j.jhydrol.2016.12.047.
Zhang, J.Q., Corlett, R.T., Zhai, D., 2019. After the rubber boom: Good news and bad news for biodiversity in Xishuangbanna, Yunnan. China. Reg. Environ. Change. 19, 1713–1724. https://doi.org/10.1007/s10113-019-01509-4.
Zhao, C.L., Jia, X.X., Zhu, Y.J., Shao, M.A., 2017. Long-term temporal variations of soil water content under different vegetation types in the Loess Plateau, China. Catena 158, 58–62. https://doi.org/10.1016/j.catena.2017.06.006.
Zhu, H., 1992. The tropical rainforest vegetation in Xishuangbanna. Chin. Geogr. Sci. 2 (1), 64–73. https://doi.org/10.1007/BF02664547.
Zhu, Q., Nie, X., Zhou, X., Liao, K., Li, H., 2014. Soil moisture response to rainfall at different topographic positions along a mixed land-use hillslope. Catena 119, 61–70. https://doi.org/10.1016/j.catena.2014.03.010.
Zhu, X., Chen, C., Wu, J., Yang, J., Zhang, W., Zou, X., Liu, W., Jiang, X., 2019. Can intercrops improve soil water infiltrability and preferential flow in rubber-based agroforestry system? Soil till Res. 191, 327–339. https://doi.org/10.1016/j.still.2019.04.017.
Zhu, X., Liu, W., Yuan, X., Chen, C., Zhu, K., Zhang, W., Yang, B., 2022. Aggregate stability and size distribution regulate rainsplash erosion: Evidence from a humid tropical soil under different land-use regimes. Geoderma 420, 115880. https://doi.org/10.1016/j.geoderma.2022.115880.
Zou, X., Zhu, X., Zhu, P., Singh, A.K., Zakari, S., Yang, B., Chen, C., Liu, W., 2021. Soil quality assessment of different *Hevea brasiliensis* plantations in tropical China. J. Environ. Manage. 285, 112147. https://doi.org/10.1016/j.jenvman.2021.112147.
Zucco, G., Brocca, L., Moramacro, T., Morbidelli, R., 2014. Influence of land use on soil moisture spatial-temporal variability and monitoring. J. Hydrol. 516, 193–199. https://doi.org/10.1016/j.jhydrol.2014.01.043.
Zuo, F.L., Li, X.Y., Yang, X.F., Ma, Y.J., Shi, F.Z., Liao, Q.W., Li, D.S., Wang, Y., Wang, R. D., 2021. Linking root traits and soil moisture redistribution under Achnatherum splendens using electrical resistivity tomography and dye experiments. Geoderma 386, 114908. https://doi.org/10.1016/j.geoderma.2020.114908.